

PID Voltage Control for Switched Reluctance Motor using Zeta Converter

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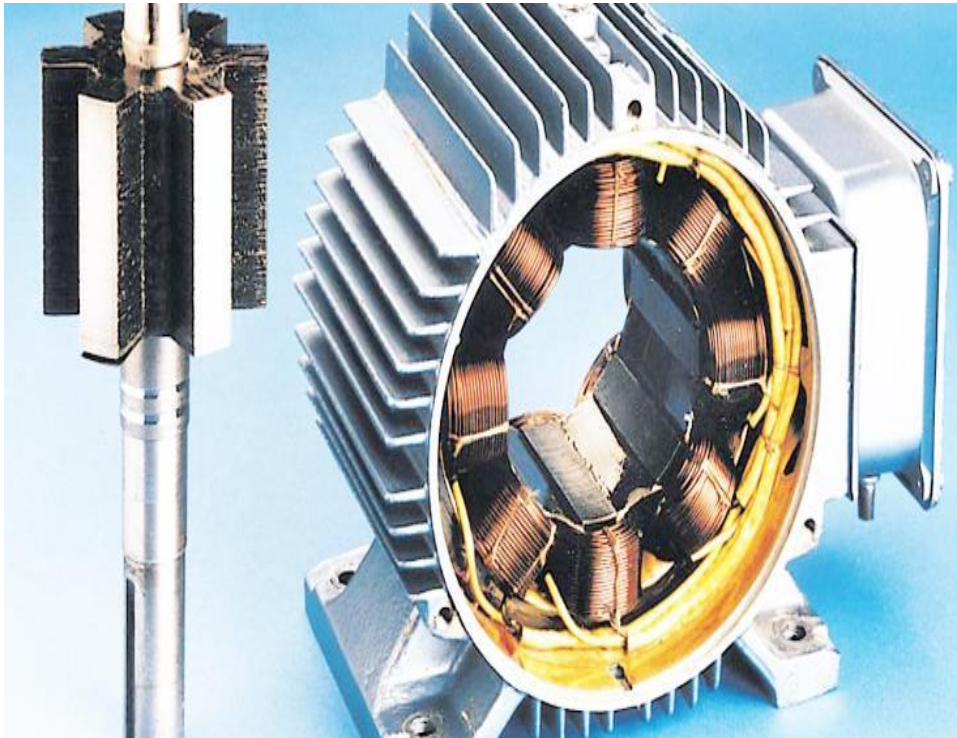


Outline

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Introduction

wide use for Switched reluctance motor(Multiple Advantages and Disadvantages for SRM : applications) :



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Objective

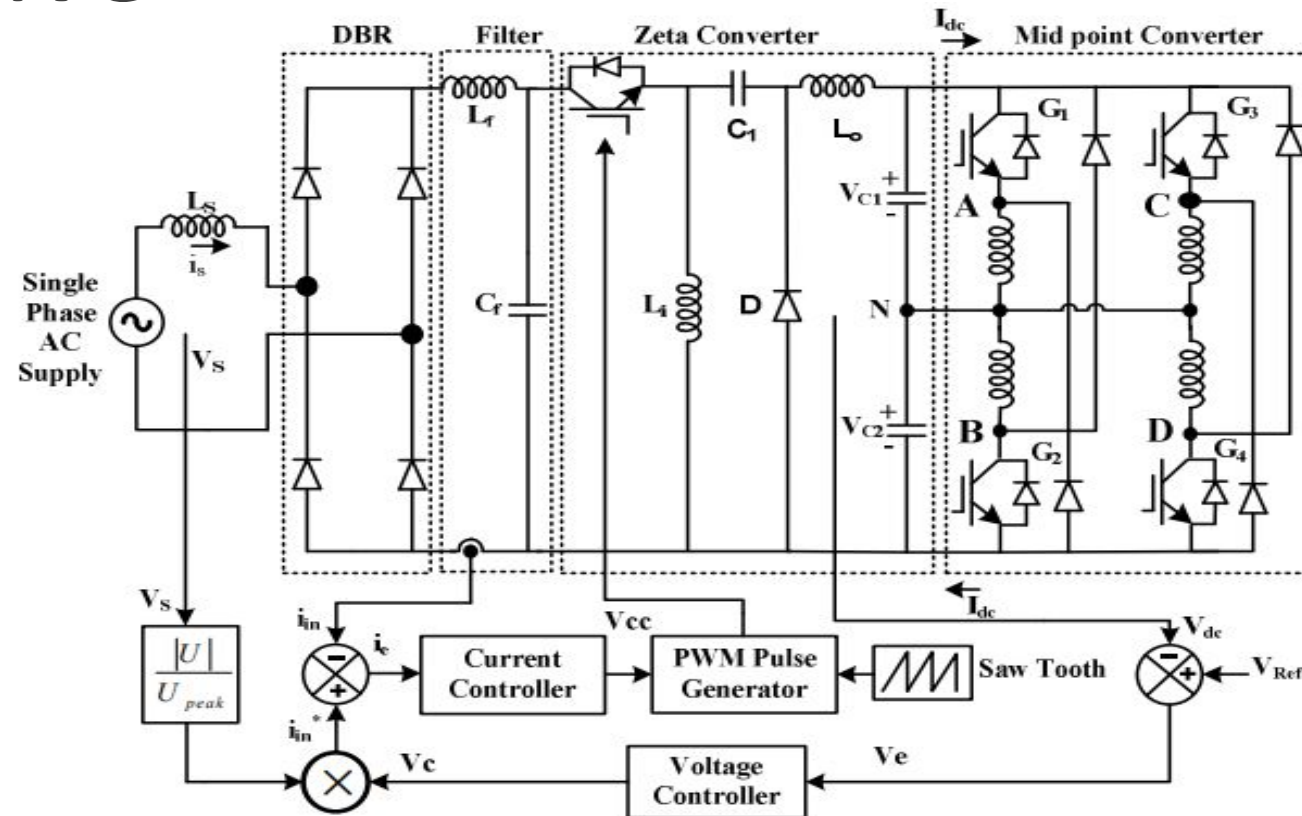


Figure 1: Switch reluctance motor fed with PFC based Zeta converter [4]



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Simulink Model (Overview)

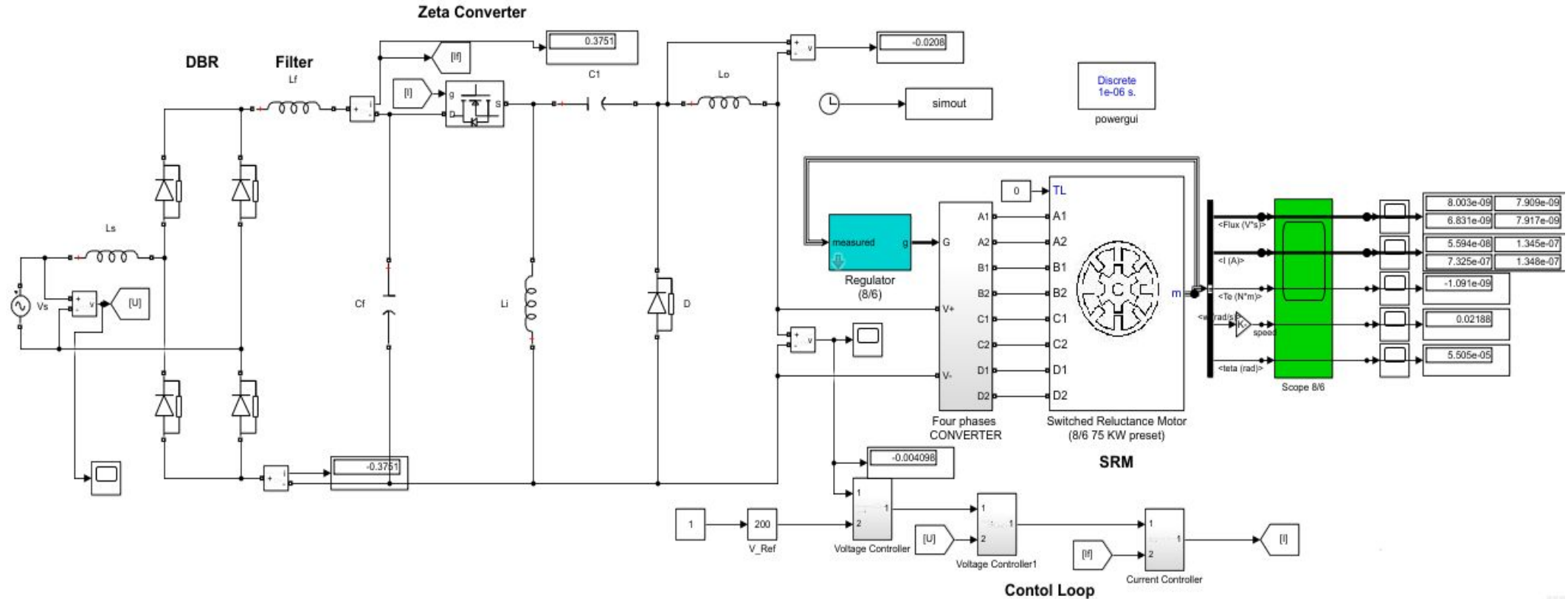


Figure 1: The project Simulink Model



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Simulink Model (ZETA Converter)

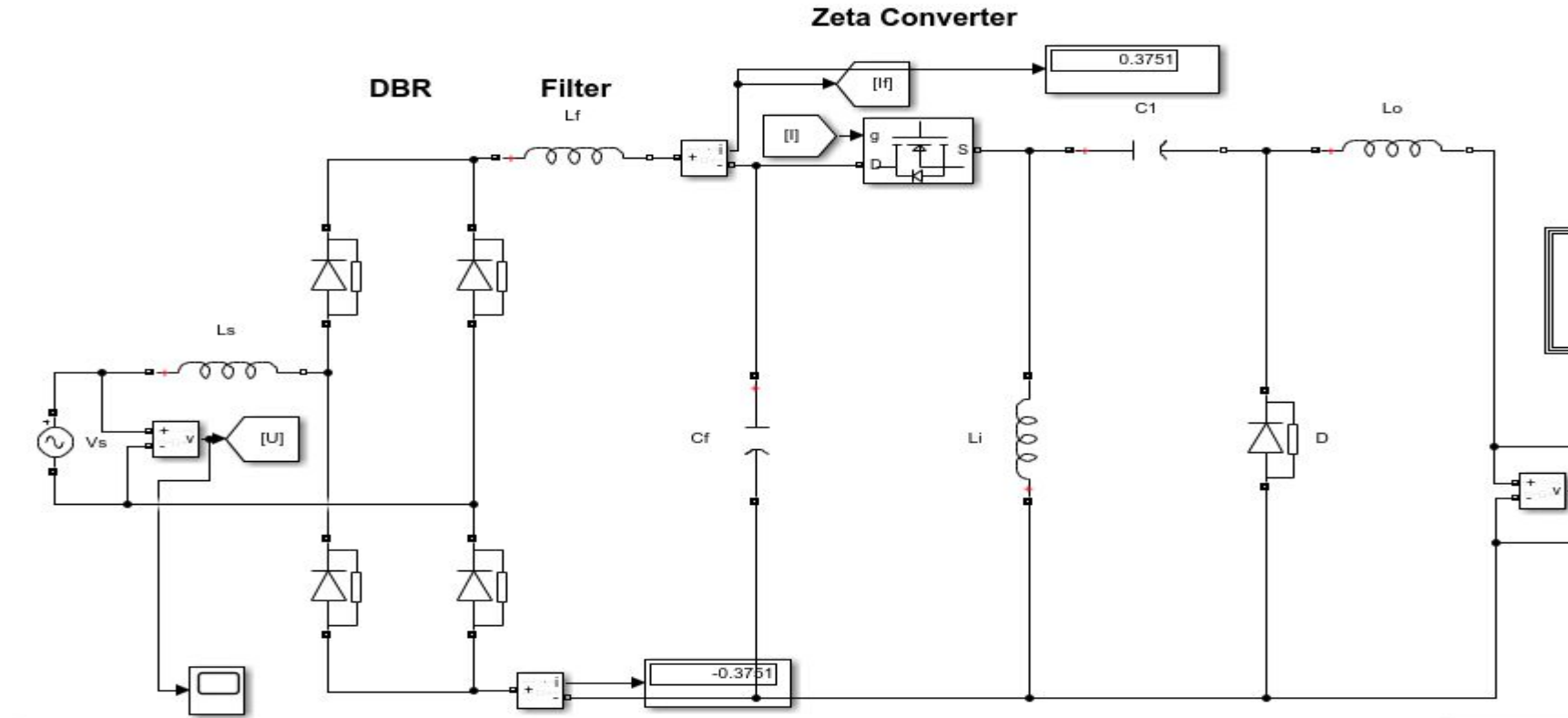


Figure 2: The ZETA Converter Simulink Model



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Simulink Model (SRM Model)

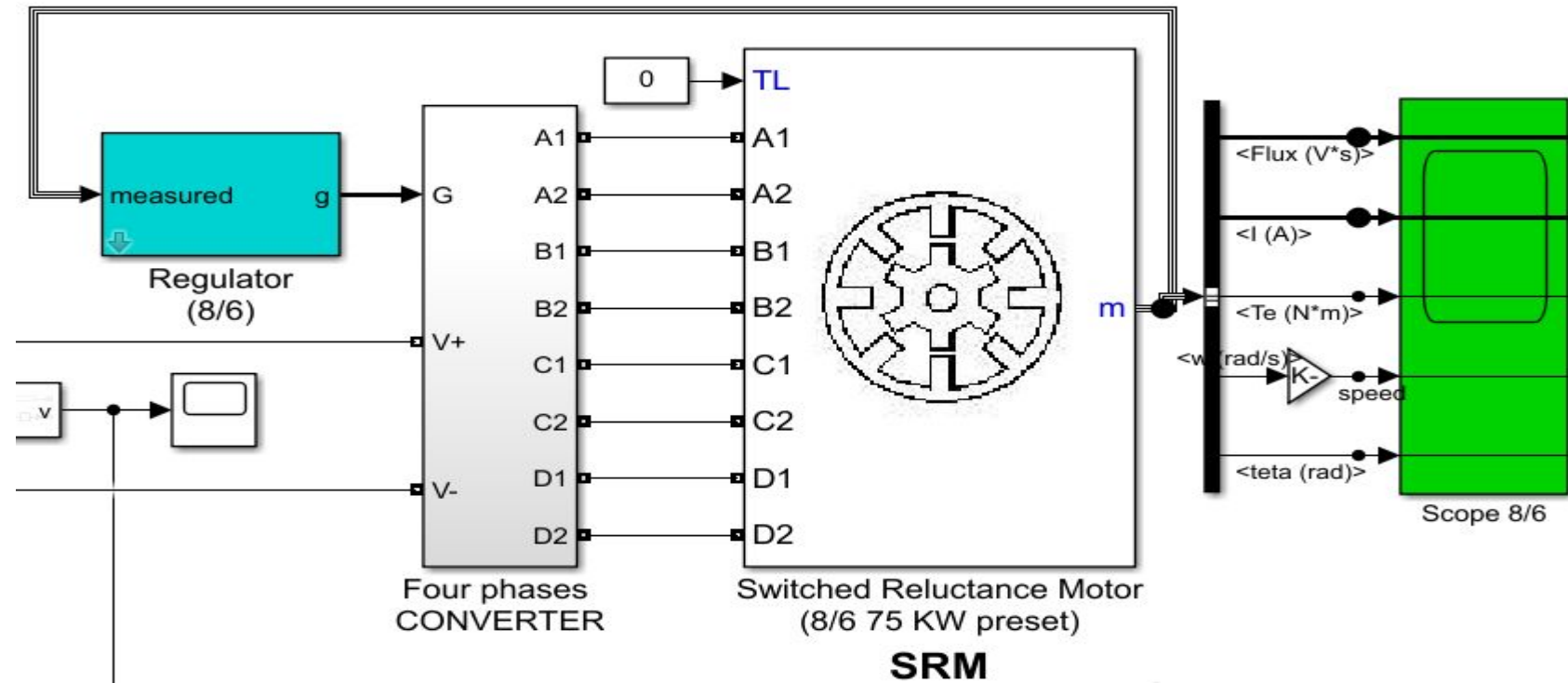


Figure 3: The SRM Simulink Model



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Simulink Model (Mid-point converter)

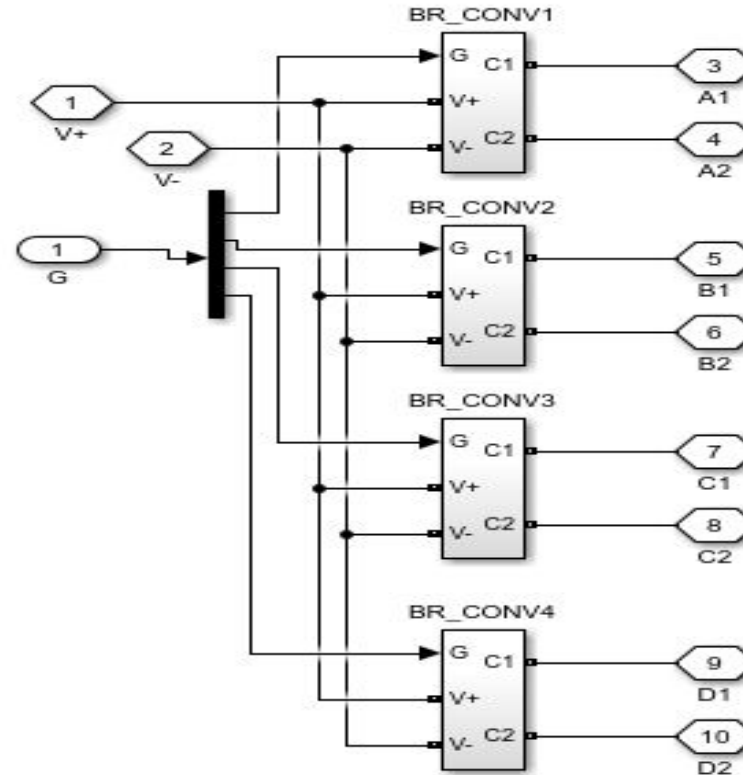


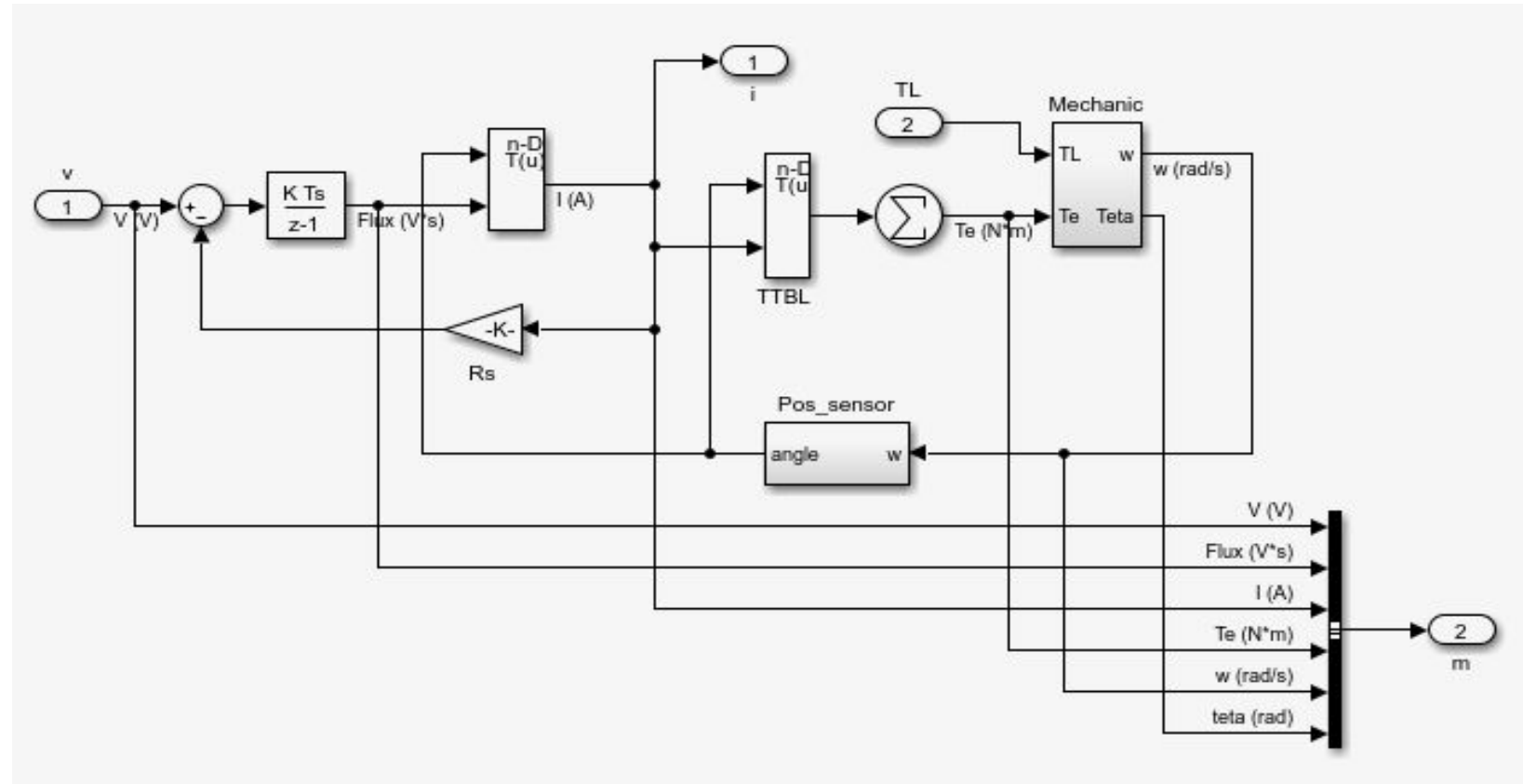
Figure 4: The Four-phase mid-point converter



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Simulink Model (Physical Model)



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Simulink Model (Control Loop)

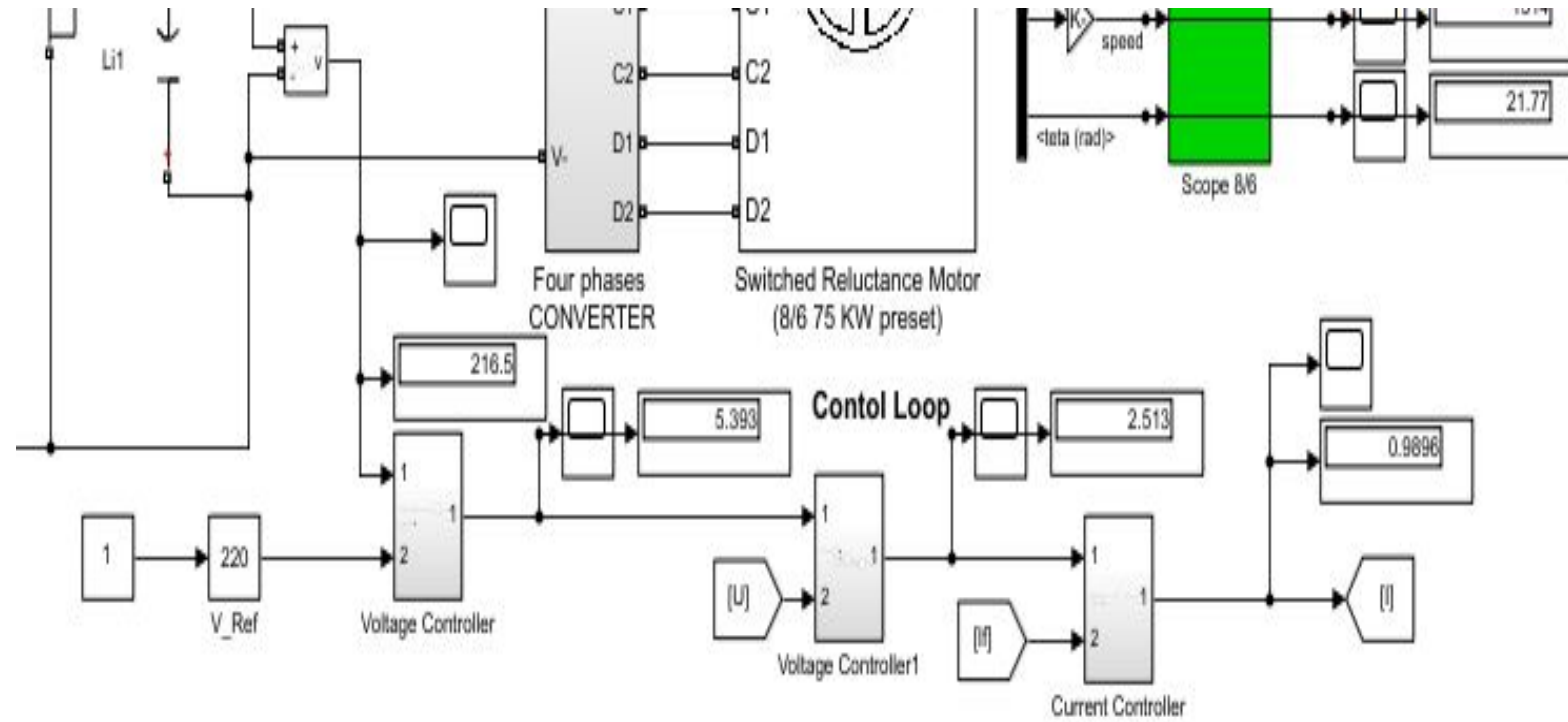


Figure 6: The PID Control loop Simulink Model



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Simulink Model (PID Controllers)

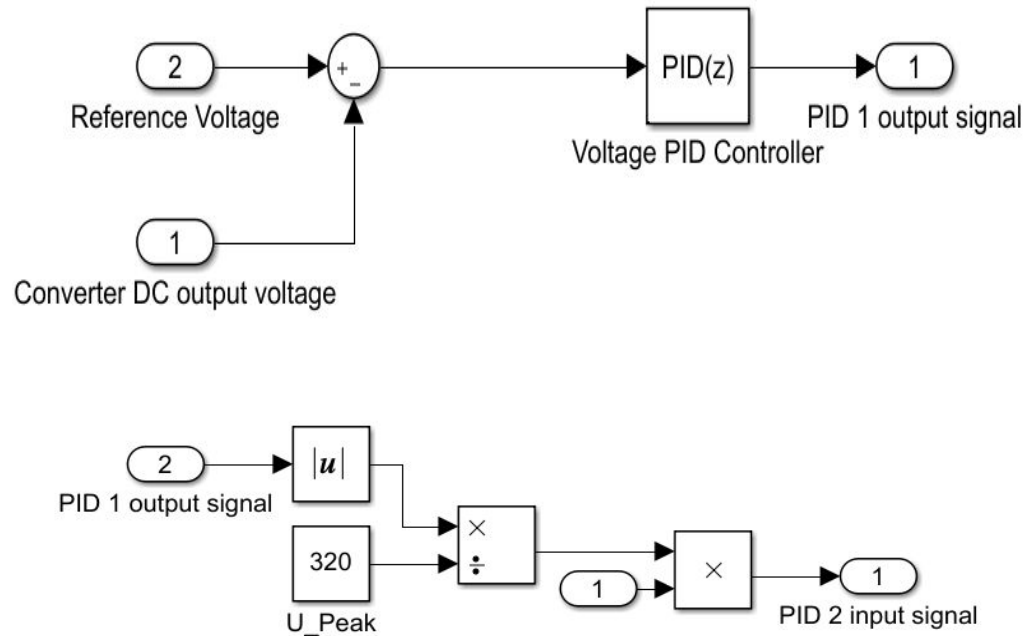


Figure 7: PID Voltage Controller

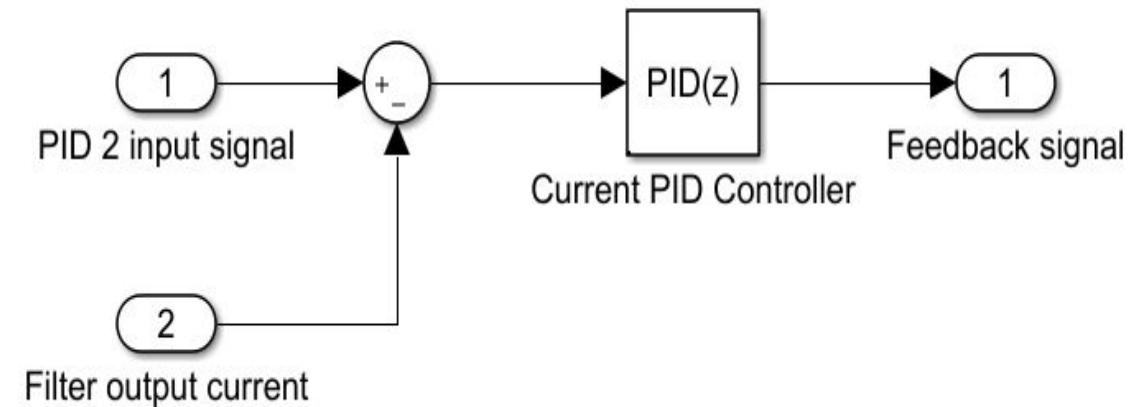


Figure 8: PID Current Controller



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Results (ZETA Converter output volt)

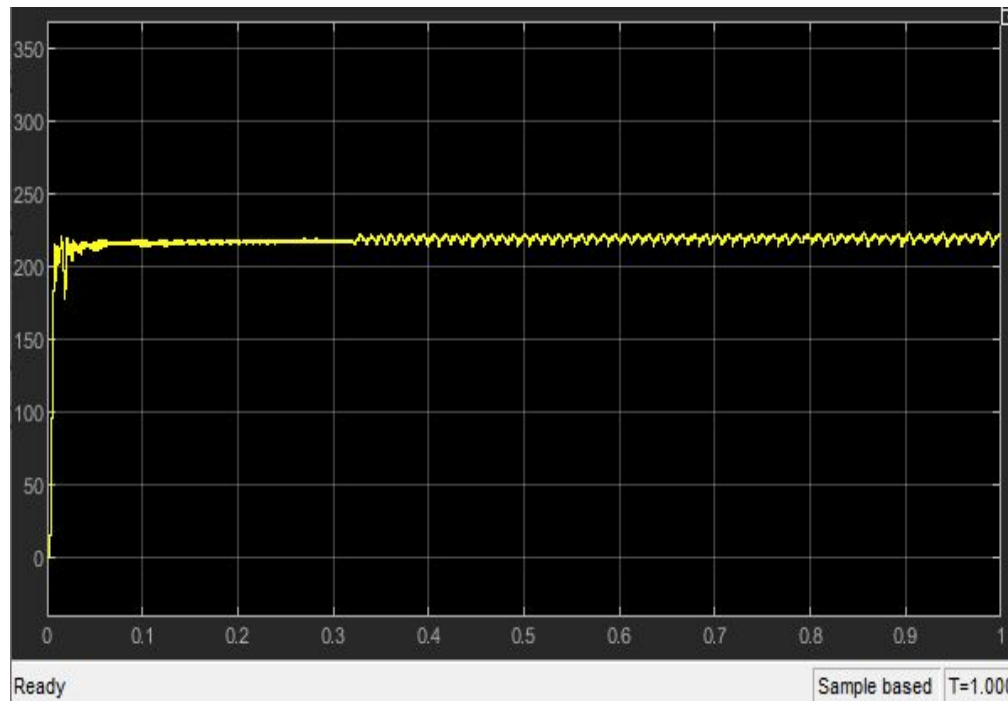


Figure 9: Converter DC output voltage at $V_{ref}=220$

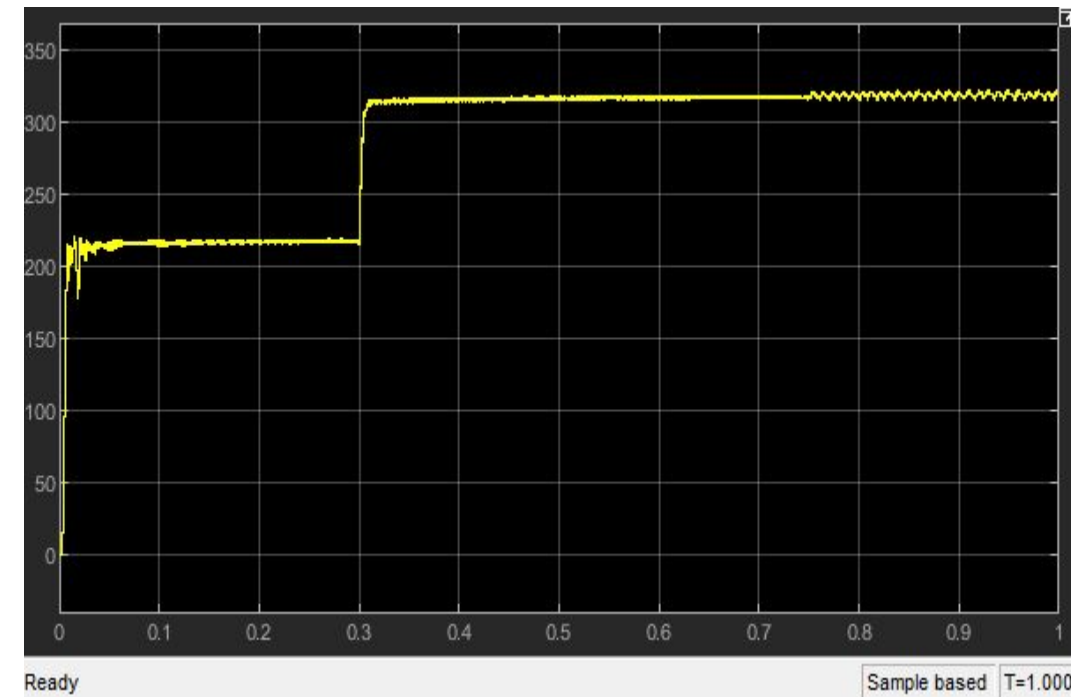


Figure 10: Converter DC output voltage at when V_{ref} changed from 220 V to 320 V



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Results (ZETA Converter output current)

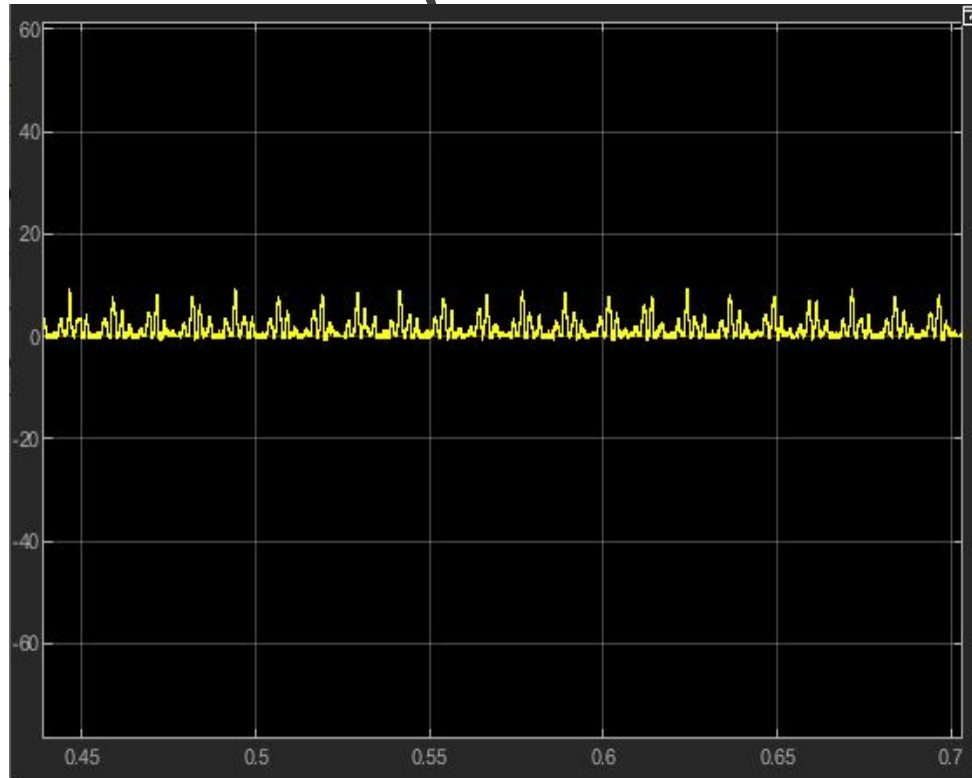


Figure 11: Converter DC current voltage at $V_{ref} = 220$

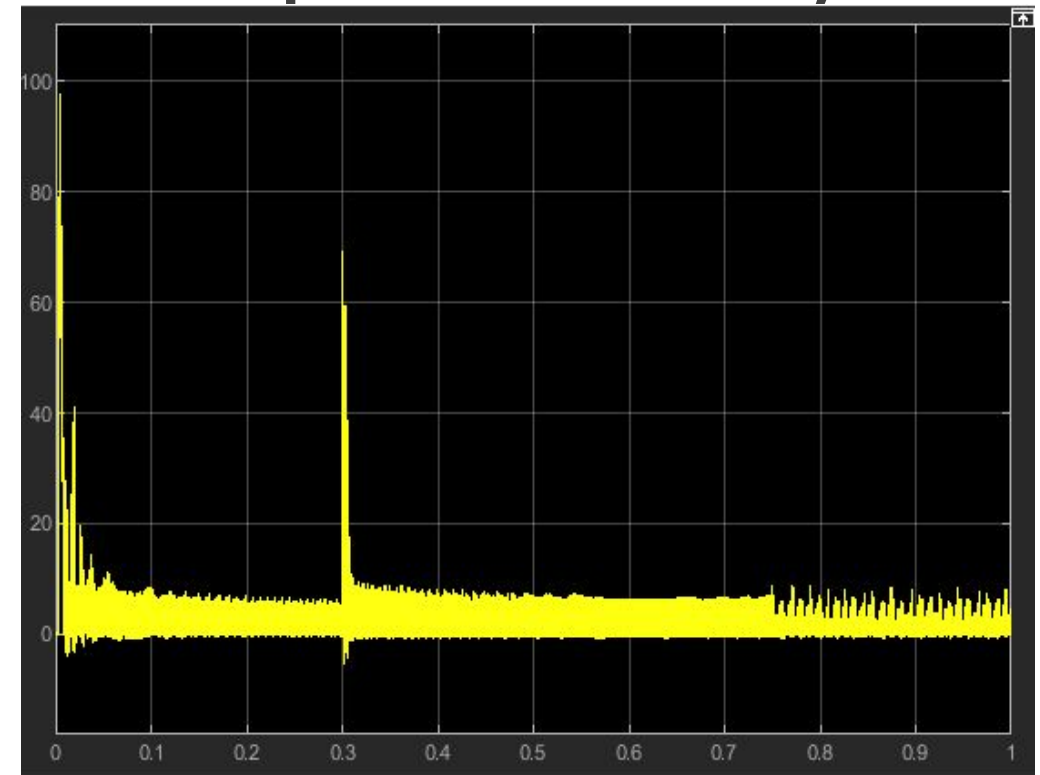


Figure 12: Converter DC output current at when V_{ref} changed from 220 V to 320 V



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Results (Motor Output Torque T_m)

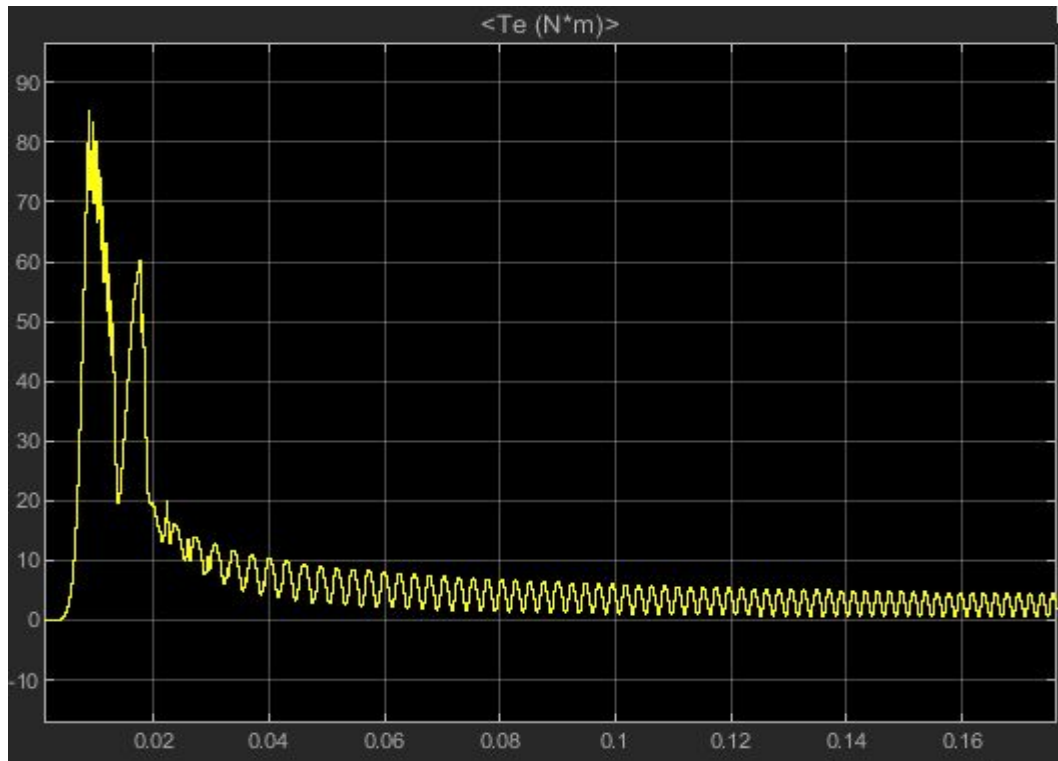


Figure 7: Motor output torque T_m

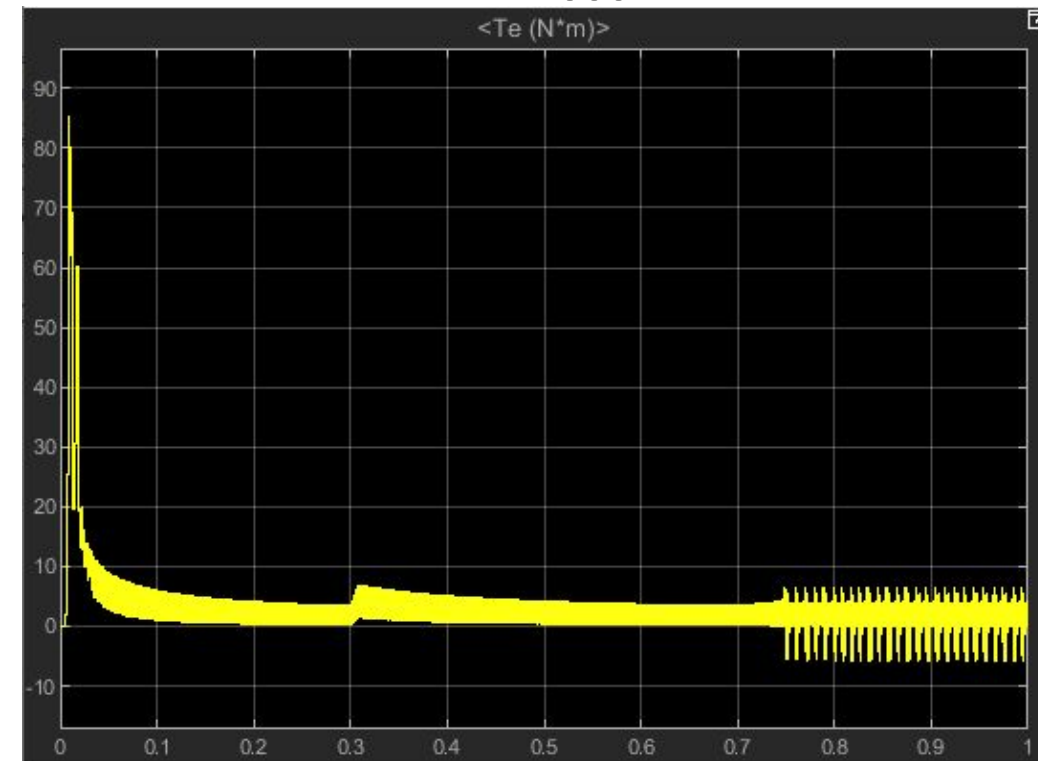


Figure 8: Motor output torque T_m when V_{ref} changed from 220V to 320V at $(t=0.3 \text{ s})$



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Results (Motor Output speed N_m)

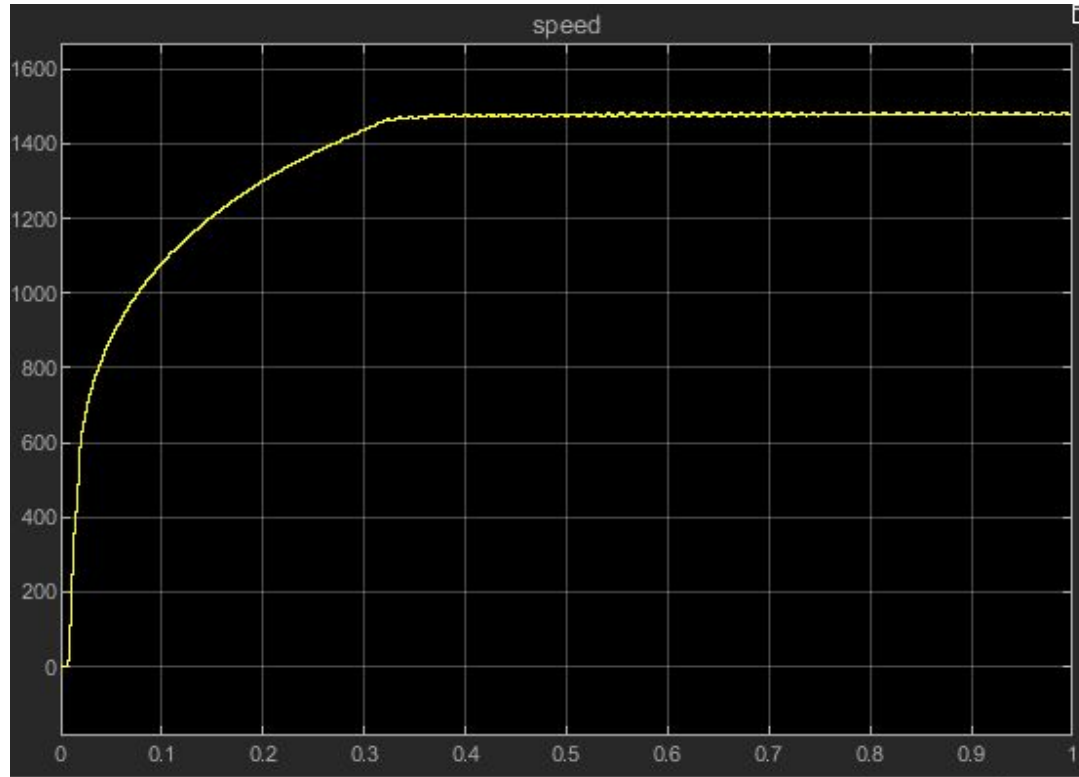


Figure 9: Motor Output speed N_M

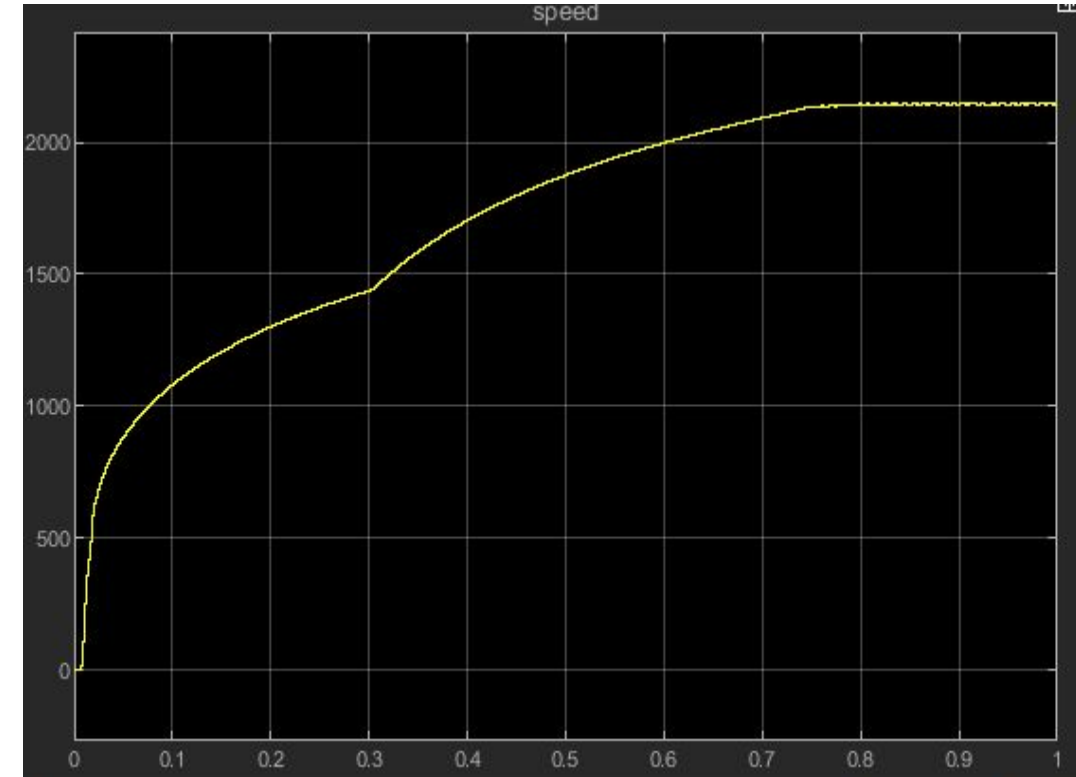


Figure 10: Motor output speed T_m when V_{ref} changed from 220V to 320V at ($t=0.3$ s)



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Results (Motor Output Braking)

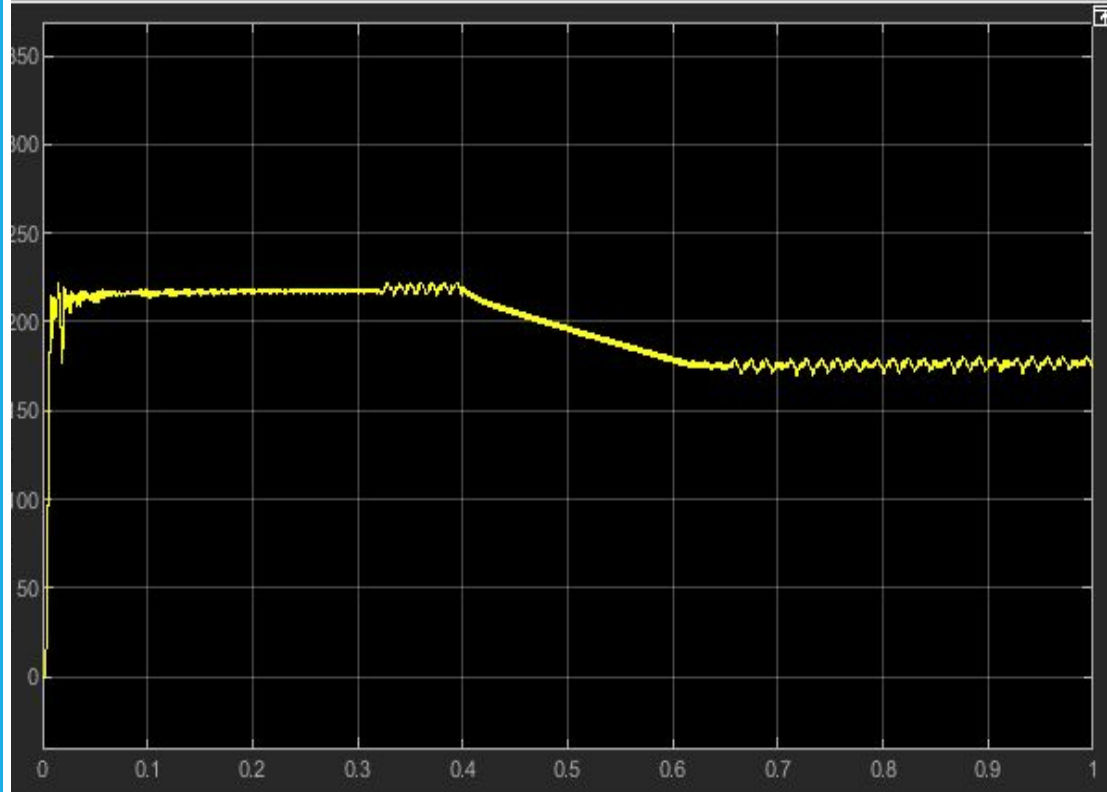


Figure 10: Converter DC output voltage at when V_{ref} changed from 220 V to 180 V

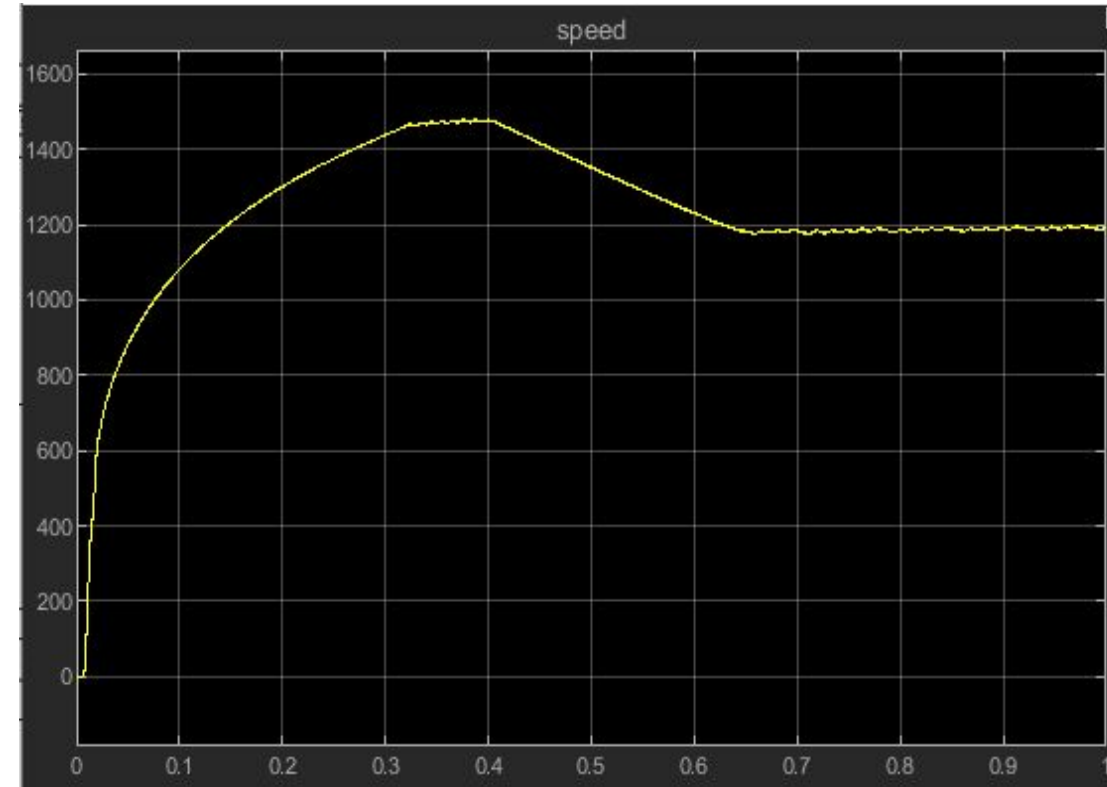


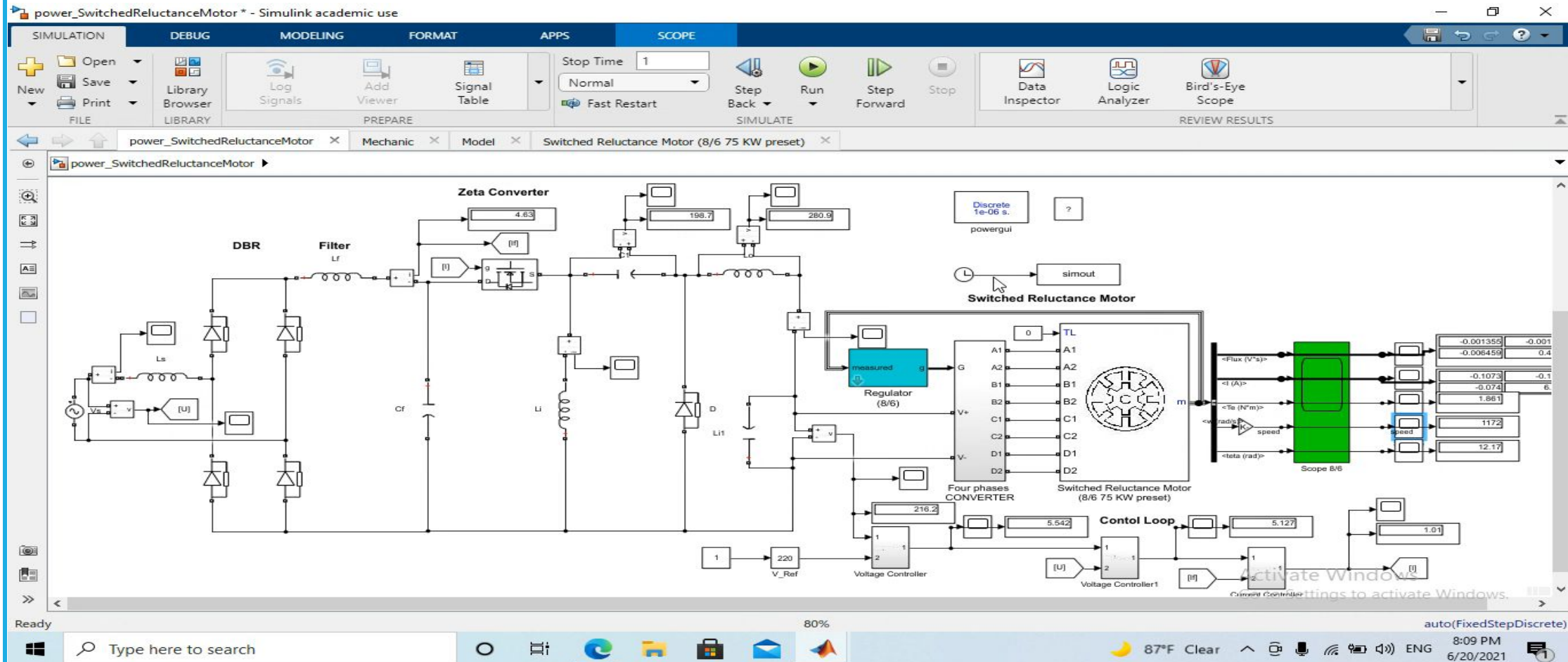
Figure 10: Motor output speed T_m when V_{ref} changed from 220V to 180V at ($t=0.4$ s)



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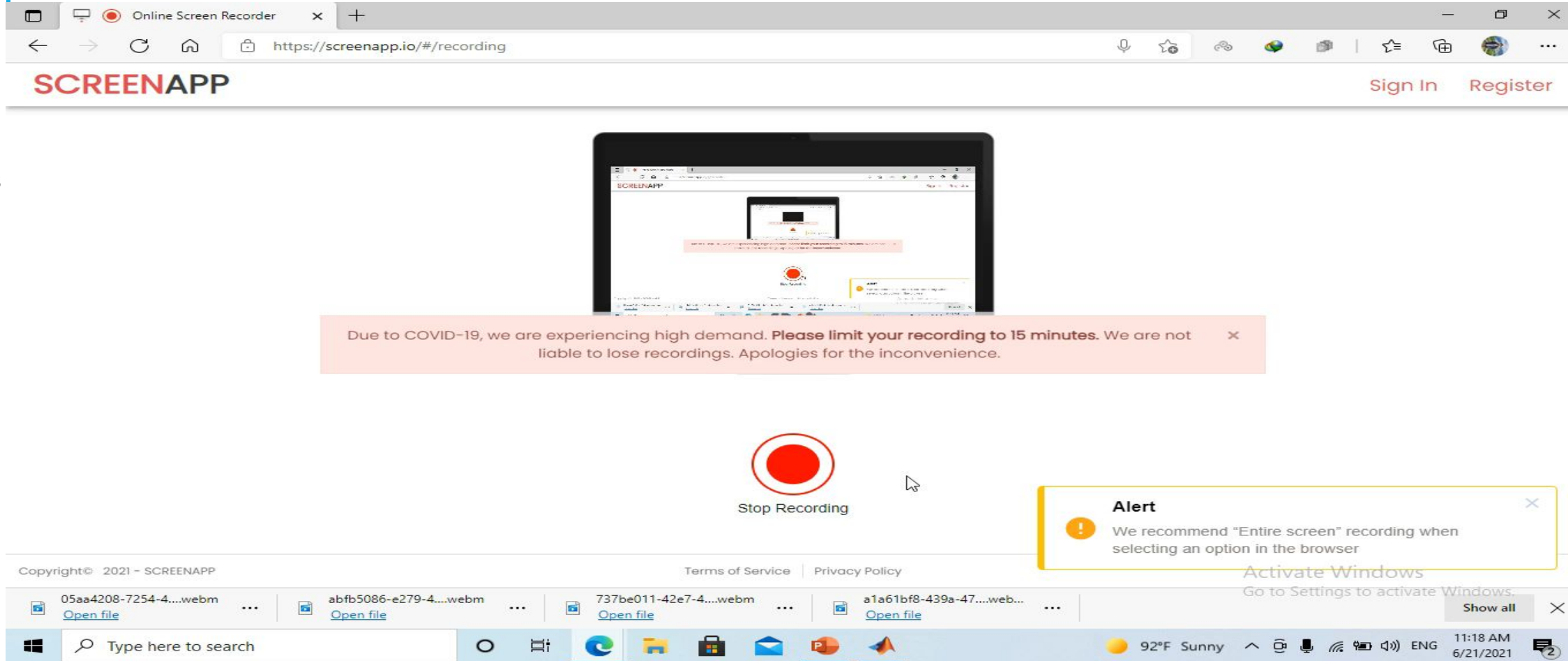
Simulation Video (Motoring)



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Simulation video (Braking)



The screenshot shows a web browser window with the URL <https://screenapp.io/#/recording>. The page features the SCREENAPP logo and navigation links for 'Sign In' and 'Register'. A central video player displays a simulation of a ZETA converter and an SRM motor. A red 'Stop Recording' button is visible below the video. A pink notification banner states: 'Due to COVID-19, we are experiencing high demand. Please limit your recording to 15 minutes. We are not liable to lose recordings. Apologies for the inconvenience.' A yellow 'Alert' box recommends 'Entire screen' recording. The Windows taskbar at the bottom shows several open webm files and the system clock indicating 11:18 AM on 6/21/2021.

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Conclusion

- A PFC based Zeta converter fed SRM motor drive has been proposed for wide range of speed.
- The PFC Zeta converter has been designed to operate in CCM and to act as an inherent power factor correction.
- The proposed SRM motor drive has been designed simulated in MATLAB/Simulink environment for achieving power factor correction at the AC mains and speed control over wide speed range.
- A satisfactory performance of drive has been achieved and the results prove it as a recommended solution for low voltage application.



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References

- [1] Schroder, G., and J. Bekiesch. "Current control for the switched reluctance motor with enhanced performance." *2005 European Conference on Power Electronics and Applications*. IEEE, 2005.
- [2] Kalan, B. A., H. C. Lovatt, and G. Prout. "Voltage control of switched reluctance machines for hybrid electric vehicles." *2002 IEEE 33rd Annual IEEE Power Electronics Specialists Conference. Proceedings (Cat. No. 02CH37289)*. Vol. 4. IEEE, 2002.
- [3] Tomczewski, Krzysztof, et al. "A Switched Reluctance Motor Drive Controller Based on an FPGA Device with a Complex PID Regulator." *Energies* 14.5 (2021): 1423.
- [4] Singh, Bhim, and Aniket Anand. "Improved power quality zeta converter fed switch reluctance motor drive." *2015 2nd International Conference on Recent Advances in Engineering & Computational Sciences (RAECS)*. IEEE, 2015.
- [5] Mathworks. Switched Reluctance Motor
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Thank you

ANY QUESTIONS ..

